DSC5001: Statistical Machine Learning I

Midterm Examination

Semester A, 2026/27

Name: _____

Student ID: _____

Total: 100 marks

- Part I contains 16 multiple-choice questions, each worth 5 marks.
- Part II contains two written questions, each worth 10 marks.
- For each multiple-choice question, select the single best answer.
- Record all multiple-choice answers in the answer table provided.
- For the written questions, show the essential reasoning and intermediate steps.
- Unless explicitly stated otherwise, use the assumptions and definitions developed in the course.

For TA Use Only

	Part I	Question 17	Question 18	Total
Maximum	80	10	10	100
Score				

Part I: Multiple-Choice Questions [80 marks]

Answer Sheet

Question	1	2	3	4	5	6	7	8
Answer								
Question	9	10	11	12	13	14	15	16
Answer								

Question 1. A disease has prevalence $P(D) = 0.05$. A diagnostic test satisfies

$$P(+ \mid D) = 0.90, \qquad P(+ \mid D^c) = 0.10.$$

What is $P(D \mid +)$?

(A) 0.050 (B) 0.321 (C) 0.900 (D) 0.947

Question 2. Suppose

$$\mathrm{Var}(X) = 4, \qquad \mathrm{Var}(Y) = 9, \qquad \mathrm{Cov}(X, Y) = -1.$$

Let $Z = 2X - Y$. What is $\mathrm{Var}(Z)$?

(A) 21 (B) 25 (C) 29 (D) 33

Question 3. Which statement is correct, assuming the relevant second moments exist?

(A) Zero covariance always implies independence.

(B) Independence implies zero covariance.

(C) Positive covariance implies that one variable is a deterministic increasing function of the other.

(D) Covariance determines the marginal variance of each variable.

Question 4. Two Bernoulli datasets are observed:

$$D_A: \text{ 3 successes in 4 trials}, \qquad D_B: \text{ 30 successes in 40 trials}.$$

Let $\ell_A(p)$ and $\ell_B(p)$ denote their log-likelihood functions. Both are maximized at $\widehat{p} = 0.75$. What is

$$\frac{|\ell_B''(0.75)|}{|\ell_A''(0.75)|}?$$

(A) 1 (B) 4 (C) 10 (D) 100

Question 5. Suppose

$$X_1, \ldots, X_4 \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$$

and the observations are $-1, 1, 3, 5$. What are the maximum-likelihood estimates of μ and σ^2?

(A) $(2, 4)$ (B) $(2, 5)$ (C) $(2, 20/3)$ (D) $(3, 5)$

Question 6. An estimator $\widehat{\theta}_n$ satisfies

$$\mathbb{E}[\widehat{\theta}_n] = \theta + \frac{1}{n}, \qquad \mathrm{Var}(\widehat{\theta}_n) = \frac{4}{n}.$$

What is its mean-squared error?

(A) $\frac{4}{n}$ (B) $\frac{1}{n^2}$ (C) $\frac{4}{n} + \frac{1}{n^2}$ (D) $\frac{4}{n} + \frac{1}{n}$

Question 7. For an estimator whose standard error behaves as

$$\mathrm{SE}(\widehat{\theta}) \propto \frac{1}{\sqrt{n}},$$

the sample size is increased from 36 to 144. Approximately how does the standard error change?

(A) It is divided by 4.

(B) It is divided by 2.

(C) It remains unchanged.

(D) It doubles.

Question 8. An observed dataset contains five distinct observations,

$$D = \{x_1, x_2, x_3, x_4, x_5\}.$$

An ordinary nonparametric bootstrap sample contains five draws with replacement. What is the probability that x_1 appears at least once in one bootstrap sample?

(A) $\left(\frac{1}{5}\right)^5$ (B) $\left(\frac{4}{5}\right)^5$ (C) $1 - \left(\frac{4}{5}\right)^5$ (D) $1 - \left(\frac{1}{5}\right)^5$

Question 9. The original dataset is fixed. The number of bootstrap repetitions is increased from $B = 1{,}000$ to $B = 10{,}000$. Which statement is correct?

(A) The larger B reduces Monte Carlo error in the bootstrap approximation, but it does not add population information to the original sample.

(B) The larger B reduces the true sampling variance of the estimator by a factor of 10.

(C) The larger B removes bias caused by an unrepresentative original sample.

(D) The larger B is equivalent to increasing the original sample size from n to $10n$.

Question 10. Let

$$\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}}$$

be the OLS residual vector. Which condition follows from the least-squares normal equations?

(A) $\boldsymbol{X}\boldsymbol{e} = \boldsymbol{0}$ (B) $\boldsymbol{X}^\top \boldsymbol{e} = \boldsymbol{0}$ (C) $\boldsymbol{y}^\top \boldsymbol{e} = 0$ (D) $\boldsymbol{e}^\top \boldsymbol{e} = 0$

Question 11. Four observations from a fitted regression model have the following residuals and leverage values:

i	1	2	3	4
e_i	2.0	1.0	0.5	3.0
h_{ii}	0.10	0.40	0.70	0.05

Using the Cook's-distance relationship developed in the course, which observation has the largest influence score, up to the common positive scaling factor?

(A) Observation 1 (B) Observation 2 (C) Observation 3 (D) Observation 4

Question 12. At a fixed new input $\boldsymbol{x}_0$, why is a prediction interval for a future response generally wider than a confidence interval for the conditional mean?

(A) The prediction interval also includes uncertainty from future observation noise.

(B) The confidence interval is computed using fewer observations.

(C) The prediction interval ignores uncertainty in the fitted coefficients.

(D) The confidence interval assumes that the fitted coefficients have zero variance.

Question 13. A new predictor is added to an OLS regression model and the enlarged model is fitted to exactly the same training observations. Which statement is necessarily true?

(A) The training RSS must increase.

(B) The training R^2 cannot decrease.

(C) The adjusted R^2 cannot decrease.

(D) The test MSE cannot increase.

Question 14. Consider the polynomial regression model

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3.$$

Which statement is correct?

(A) The model is nonlinear in the coefficients and therefore cannot be fitted by ordinary least squares.

(B) The model is linear in x because it is linear in the coefficients.

(C) The model is nonlinear in x but linear in the coefficients, so it can be fitted as linear regression on transformed features.

(D) The model can only be fitted if $x > 0$.

Question 15. A cubic regression spline uses the basis

$$\boldsymbol{\phi}(x) = \left[1, x, x^2, x^3, (x-1)_+^3, (x-3)_+^3\right]^\top.$$

What is $\boldsymbol{\phi}(4)$?

(A) $[1, 4, 16, 64, 27, 1]^\top$

(B) $[1, 4, 16, 64, 3, 1]^\top$

(C) $[1, 4, 16, 64, 27, 0]^\top$

(D) $[1, 4, 16, 64, 1, 27]^\top$

Question 16. Four polynomial regression models give the following training and validation errors:

Degree	1	3	5	9
Training MSE	6.2	3.1	1.5	0.4
Validation MSE	6.5	3.4	2.6	3.8

Which polynomial degree should be selected using the validation criterion developed in the course?

(A) Degree 1 (B) Degree 3 (C) Degree 5 (D) Degree 9

Question 17: Quadratic Regression [10 marks]

Consider the five observations

$$x = (-2, -1, 0, 1, 2), \qquad y = (4, 1, 1, 3, 6).$$

A quadratic regression model with an intercept is fitted:

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2.$$

All parts below refer to this model. Do not use statistical software.

(a) Write the transformed feature vector $\boldsymbol{\phi}(x)$ and construct the complete design matrix $\boldsymbol{X}$ for the five observations. [1 mark]

(b) Compute $\boldsymbol{X}^\top \boldsymbol{X}$. [1 mark]

(c) Compute $\boldsymbol{X}^\top \boldsymbol{y}$. [1 mark]

(d) Use the normal equations to obtain $\widehat{\beta}_0$, $\widehat{\beta}_1$, and $\widehat{\beta}_2$, and write the fitted quadratic function explicitly. [2 marks]

(e) Compute the fitted value $\widehat{y}_i$ and residual $e_i = y_i - \widehat{y}_i$ for each of the five observations. [2 marks]

(f) Compute the residual sum of squares RSS and the training MSE $RSS/5$. [1 mark]

(g) Use the fitted model to predict the mean response at $x_0 = 1.5$. Then explain in one or two sentences why this is a nonlinear regression model even though its coefficients can be estimated by ordinary linear least squares. [2 marks]

Question 18: Bias–Variance Decomposition [10 marks]

Fix an input value x_0. A future response is generated by

$$Y_0 = f(x_0) + \varepsilon_0,$$

where

$$\mathbb{E}[\varepsilon_0] = 0, \qquad \mathrm{Var}(\varepsilon_0) = \sigma^2.$$

A learning procedure is fitted on a random training dataset D and produces the prediction $\widehat{f}_D(x_0)$. Assume that the future noise ε_0 is independent of D.

Define

$$m(x_0) = \mathbb{E}_D[\widehat{f}_D(x_0)].$$

The quantity of interest is

$$\mathbb{E}_{D,\varepsilon_0}\left[\left(Y_0 - \widehat{f}_D(x_0)\right)^2\right].$$

Do not quote the final bias–variance decomposition without showing the intermediate steps.

(a) Define the bias of $\widehat{f}_D(x_0)$ in terms of $f(x_0)$ and $m(x_0)$. [1 mark]

(b) Substitute $Y_0 = f(x_0) + \varepsilon_0$ into the prediction error and add and subtract $m(x_0)$. Express $Y_0 - \widehat{f}_D(x_0)$ as the sum of three terms. [2 marks]

(c) Square the expression from part (b). Identify the three squared terms and the three cross terms. [2 marks]

(d) Take expectation over both D and ε_0. Explain why each cross term has expectation zero. [2 marks]

(e) Derive the final decomposition of the expected squared prediction error and identify the irreducible-noise, squared-bias, and variance components. [2 marks]

(f) Suppose model flexibility is gradually increased. Use the decomposition to explain why test error may first decrease and then increase even while training error continues to decrease. [1 mark]